# Acoustic systems (5th class)

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Department of Electronic Engineering

- Piston source
- Radiation impedance
- Sound pressure level

# Measure for sound visualization

# Visualization of sound propagation

Generally, visualization of sound is difficult at audible range.

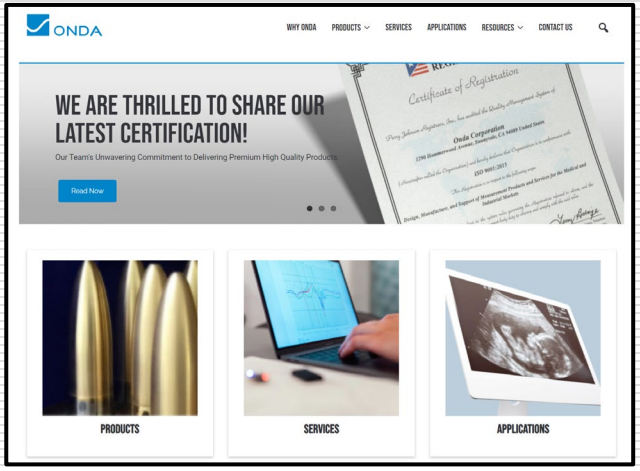
Ultrasonic frequency range ( over 20kHz)

- **Schlieren Photography**
- **Opto-elastic method**
- **PIV(Particle imaging velocimetry)**

**and so on**

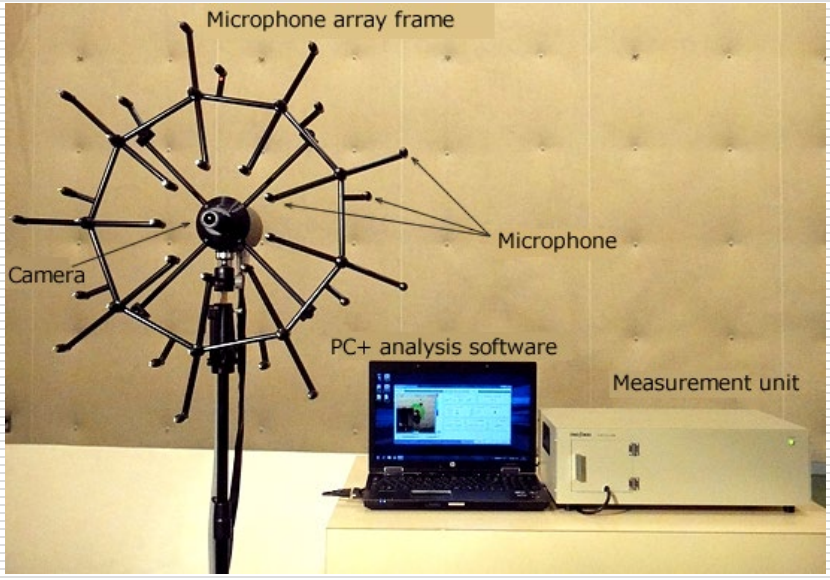



From ONDA Web page (US)
https://www.ondacorp.com/



# Microphone array

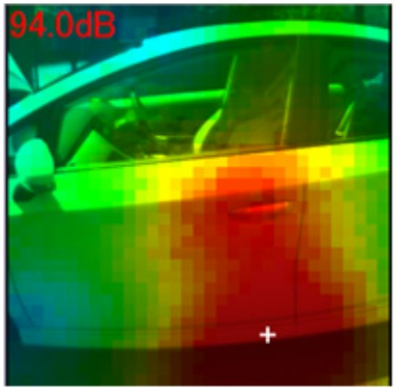
Microphone array is employed for visualization.







Measurement example2:
operating sound when printing




Measurement example3:
sound when closing door

https://www.onosokki.co.jp/English/hp_e/products/application2/beamforming.htm

From Onosokki web page

# Sound field emitted from point source

# Sound wave from point source: sound pressure $p$

In the absence of reflection—that is, when only a forward-traveling wave exists—the velocity potential is given by

$$\Phi(x,t) = \frac{C_1}{r} e^{j(\omega t - kr)}$$

Generally, only forward wave propagates from the spherical point source.

The Sound pressure $p$ can be obtained by taking the time derivative of the velocity potential, as follows:

$$p = \rho \frac{\partial \Phi}{\partial t} = j\omega\rho\Phi$$

# Sound wave from point source: particle velocity $v$

Particle velocity $v$ can be obtained by the spatial derivative of the velocity potential as follows:

$$v = -\frac{\partial \Phi}{\partial r} = -\frac{\partial}{\partial r}\left(\frac{C_1}{r} e^{j(\omega t - kr)}\right) = -\frac{\partial}{\partial r}\left(\frac{C_1}{r}\right) e^{j(\omega t - kr)} - \frac{C_1}{r}\frac{\partial\left(e^{j(\omega t - kr)}\right)}{\partial r}$$

$$= -\left(-\frac{C_1}{r^2} e^{j(\omega t - kr)}\right) - \frac{C_1}{r}\left(-jkr\right) e^{j(\omega t - kr)} = \frac{C_1}{r^2}\left(1 + jkr\right) e^{j(\omega t - kr)}$$

$$= \frac{1 + jkr}{r}\Phi$$

# Specific acoustic impedance

sound pressure $\boldsymbol{p}$ : $p = \rho \dfrac{\partial \Phi}{\partial t} = j\omega\rho\Phi$

particle velocity $\boldsymbol{v}$ : $\boldsymbol{v} = -\boldsymbol{\nabla}\Phi = \dfrac{1+jkr}{r}\Phi$

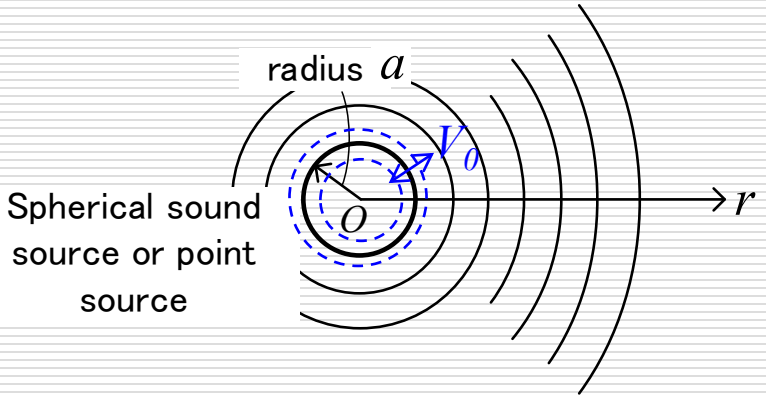
$$\frac{\boldsymbol{p}}{\boldsymbol{v}} = \frac{j\omega\rho r}{1+jkr} = \rho c \frac{jkr}{1+jkr} \qquad \text{by } \omega = kc$$

The values $p/v$ in the sound field is called by "specific acoustic impedance". Generally, specific acoustic impedance is expressed by $Z_S$.

# Volume velocity

Particle velocity $V_0$ at the surface of the spherical sound source is as follows,

$$V_0 = -\left.\frac{\partial \Phi}{\partial r}\right|_{r=a} = \frac{C_1}{a^2}\left(1 + jka\right)e^{j(\omega t - ka)}$$




When the sound source almost can be treated as a point source, that is $a << \lambda$, particle velocity is re-written as $V_0 = C_1/a^2$, and velocity potential is as follows

$$\Phi = \frac{V_0 a^2}{r} e^{j(\omega t - kr)} = \frac{4\pi a^2 V_0}{4\pi r} e^{j(\omega t - kr)}$$

$Q = 4\pi a^2 V_0$ is called as "volume velocity".

# Sound field emitted by point source

Velocity potential and sound pressure are re-written as follows,

$$\Phi = \frac{Q}{4\pi r} e^{j(\omega t - kr)} \quad , \qquad p = \rho \frac{\partial \Phi}{\partial t} = j\omega\rho \frac{Q}{4\pi r} e^{j(\omega t - kr)}$$

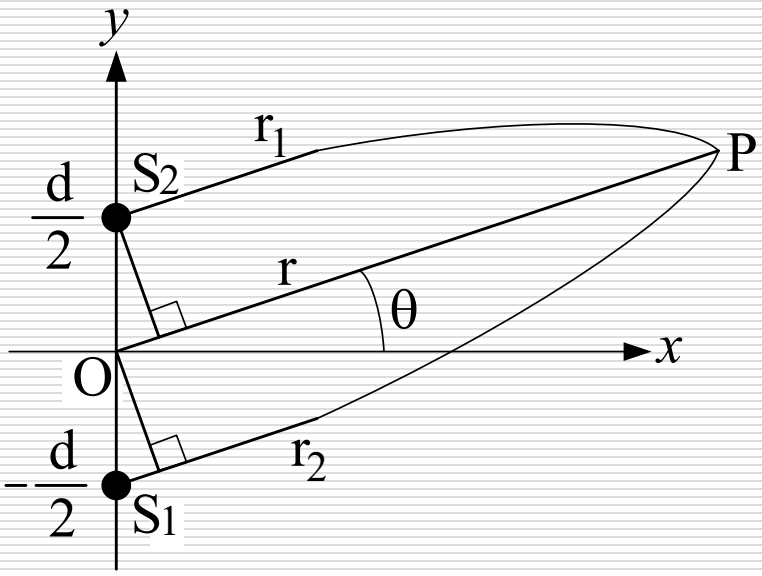
Point source creates diffusion sound field where sound pressure becomes a tenth when the distance from the source becomes ten times.

Sound source like point source is called as non-directional sound source, or as omni-directional sound source.

# Directivity index

# Sound field emitted by two point sources

When two separate point sources are emitted in phase as shown in the right, the velocity potential at P is given by followings.




$$\Phi = \left( \frac{Q}{4\pi r_1} e^{-jkr_1} + \frac{Q}{4\pi r_2} e^{jkr_2} \right) e^{j\omega t}$$

# Sound field emitted by two point sources at far field.

In the far field from the source, r >> d, $r_1 \approx r_2 \approx r$ and the velocity potential is rewritten as follows.

$$\Phi = \left( \frac{Q}{4\pi r} e^{-jk\left(r + \frac{kd}{2}\sin\theta\right)} + \frac{Q}{4\pi r} e^{-jk\left(r - \frac{kd}{2}\sin\theta\right)} \right) e^{j\omega t}$$

$$\Phi = \frac{Q}{2\pi r} e^{j(\omega t - kr)} \cos\left(\frac{kd}{2}\sin\theta\right)$$

# Derivation of directivity index

From Euler's formula: $e^{jkr} = \cos(kr) + j\sin(kr)$

From eq.(2): $$\Phi = \left( \frac{Q}{4\pi r} e^{-jk\left(r+\frac{kd}{2}\sin\theta\right)} + \frac{Q}{4\pi r} e^{-jk\left(r-\frac{kd}{2}\sin\theta\right)} \right) e^{j\omega t}$$

$$\Phi = \frac{Qe^{j\omega t}}{4\pi r}\left\{\cos\left(k\left(r+\frac{kd}{2}\sin\theta\right)\right) - j\sin\left(k\left(r+\frac{kd}{2}\sin\theta\right)\right)\right\}$$

$$+\frac{Qe^{j\omega t}}{4\pi r}\left\{\cos\left(k\left(r+\frac{kd}{2}\sin\theta\right)\right) + j\sin\left(k\left(r+\frac{kd}{2}\sin\theta\right)\right)\right\}$$



Eq.(1) is obtained. $$\Phi = \frac{Q}{2\pi r} e^{j(\omega t - kr)} \cos\left(\frac{kd}{2}\sin\theta\right)$$

# Directivity index

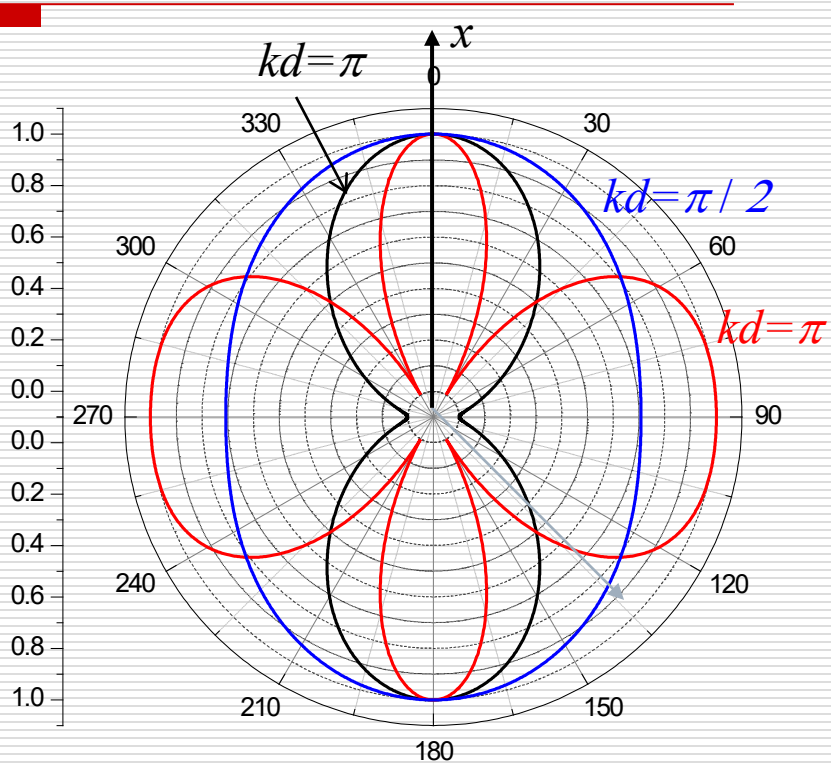
Directivity index $D(\theta)$ written by following indicates the sound pressure distribution in the same distance from sound source as a function of direction angle $\theta$.

$$D(\theta) = \left| \cos\left( \frac{k\,d}{2} \sin\theta \right) \right|$$

where maximum sound pressure is 1 by normalization.

# Directional pattern of two point sources

Right figure shows the sound pressure distribution along direction angle $\theta$ at $kd=\pi$ and $kd=\pi/2$.

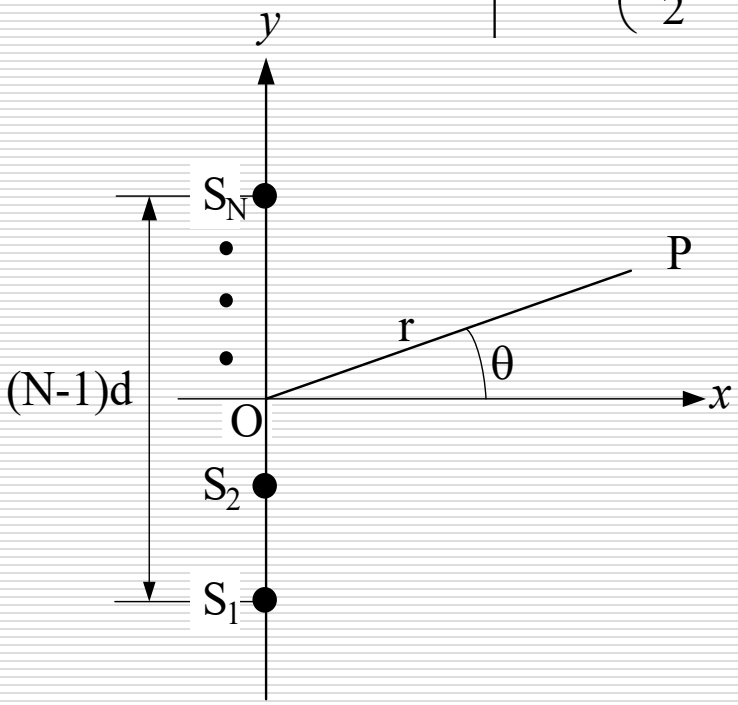


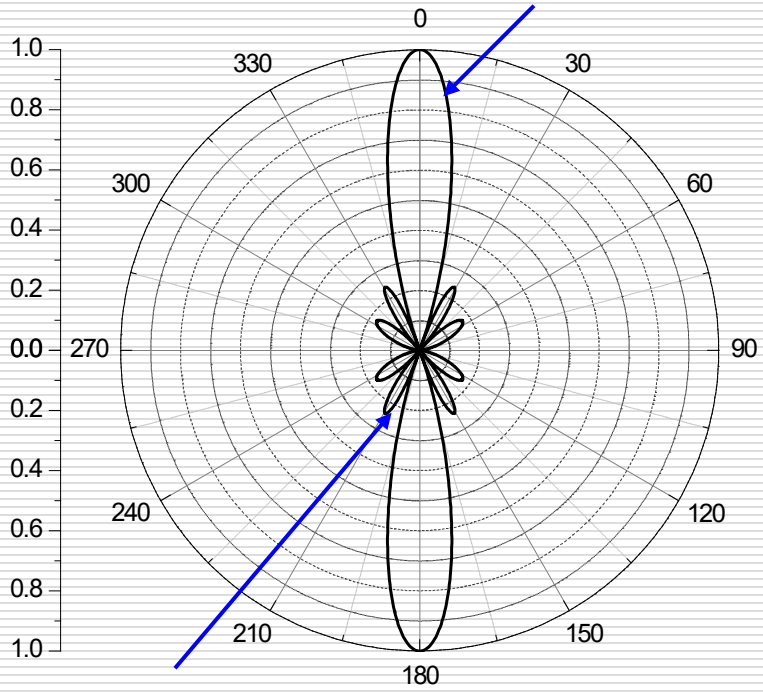

Right figure is called as "directional pattern".

# Directional pattern of multiple point sources

Directivity index of multiple point sources is given by

$$D(\theta) = \left| \frac{\sin\left(\frac{N k d}{2}\sin\theta\right)}{N \sin\left(\frac{k d}{2}\sin\theta\right)} \right|$$



# (Previous) Question 1

Draw the directivity patterns on the graph respectively when the distance of the two sound and frequency are following. In drawing, the sound velocity is 340m/s.

(a) Frequency $f$ : 680Hz,  Distance $d$ : 25cm

(b) Frequency $f$ : 340Hz,  Distance $d$ : 25cm

# (Previous) Question 1: (Answer)

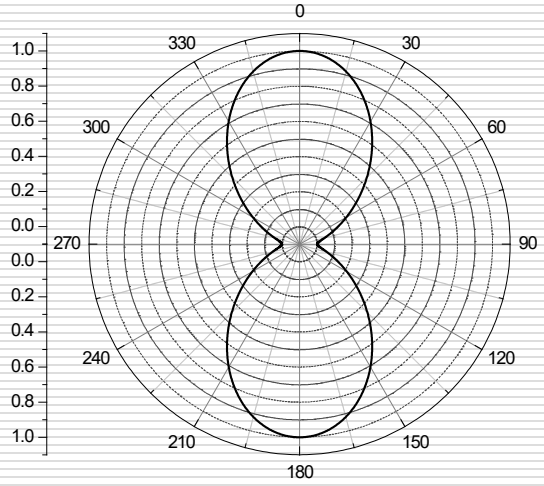
(a) Frequency $f$ : 680Hz, Distance $d$ : 25cm

$$kd = \pi, D(\theta) = \left|\cos\left(\frac{kd}{2}\sin(\theta)\right)\right| = \left|\cos\left(\frac{\pi}{2}\sin(\theta)\right)\right|$$

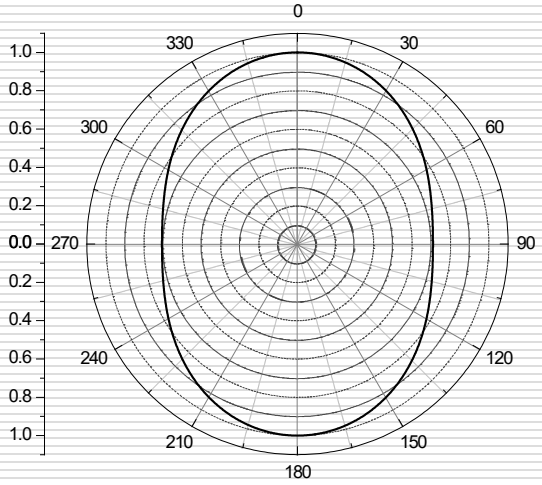
(b) Frequency $f$ : 340Hz, Distance $d$ : 25cm

$$kd = \frac{\pi}{2}, D(\theta) = \left|\cos\left(\frac{kd}{2}\sin(\theta)\right)\right| = \left|\cos\left(\frac{\pi}{4}\sin(\theta)\right)\right|$$

(a)



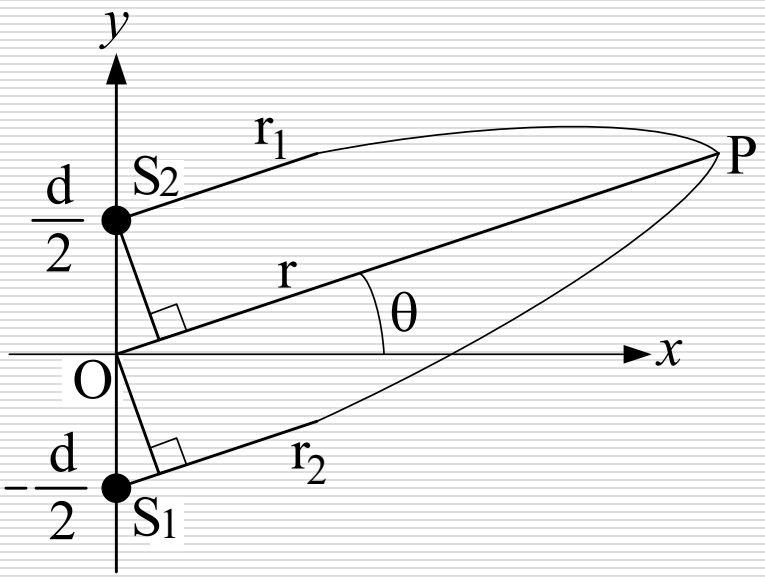

(b)



# Sound field formed by doublet source

# Sound field emitted by doublet source

When two sources $S_1$ and $S_2$ emit sound wave in antiphase, we call "doublet source", the sound pressure is expressed by the following.




$$\Phi = j\frac{Q}{2\pi r}e^{j(\omega t - kr)}\sin\left(\frac{kd}{2}\sin\theta\right)$$

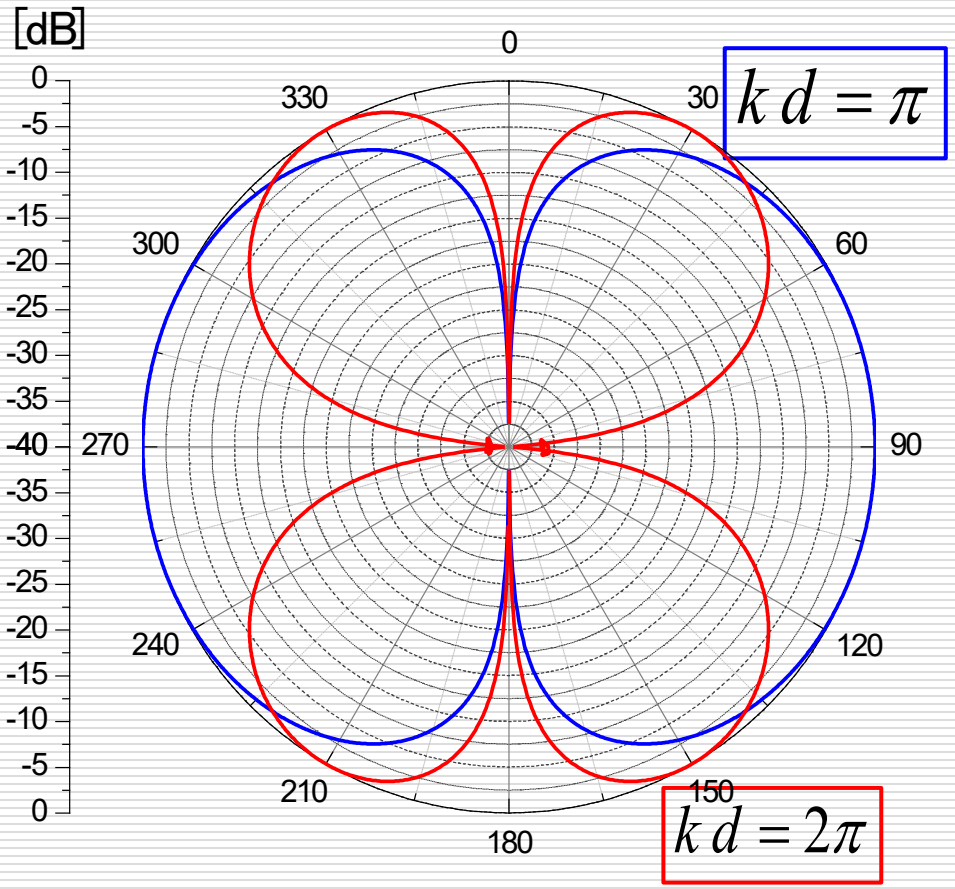
At the condition, the directivity is as follows.

$$D(\theta) = \left|\sin\left(\frac{kd}{2}\sin\theta\right)\right| \quad (kd << 1)$$

# Directional pattern of doublet source

Right figure shows the sound pressure distribution along direction angle $\theta$ at $kd=\pi$ and $kd=2\pi$.

The front of the double source is always minimum.



# Sound pressure level

# Sound pressure level

It is appropriate to treat sound pressure level logarithmically.

Sound pressure is expressed by the following.
Sound Pressure Level [S.P.L.]

$$\text{S.P.L.} = 20\, log_{10}\left(\frac{p}{p_{ref}}\right)\ [\text{dB}]$$

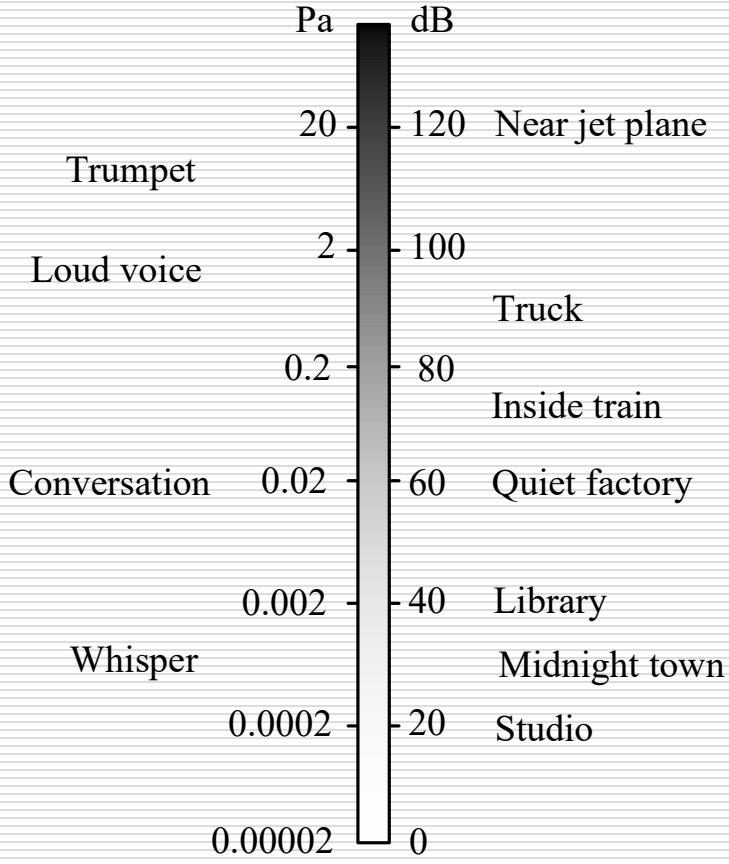
where reference is $p_{ref} = 20\ [\mu Pa]$

$p_{ref}$ corresponds to the minimum sound pressure which the people can hear.

# Example of S.P.L. in the life

Example S.P.L. against several situations.

Please calculate the particle velocity for each S.P.L.

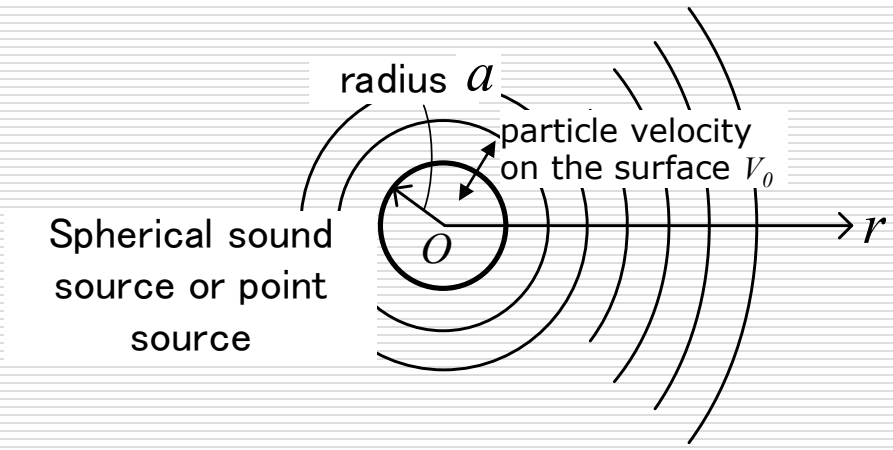


# Radiation impedance

# Sound field of the point source

Velocity potential of the point source:

$$\Phi = \frac{V_0 S}{2\pi r} e^{j(\omega t - kr)} = \frac{Q}{2\pi r} e^{j(\omega t - kr)}$$




$Q$ is volume velocity.

sound pressure:

$$p = \rho \frac{\partial \Phi}{\partial t}$$

Velocity potential:

$$v = -\frac{\partial \Phi}{\partial r}$$

# The relation between characteristic acoustic impedance and specific acoustic impedance.

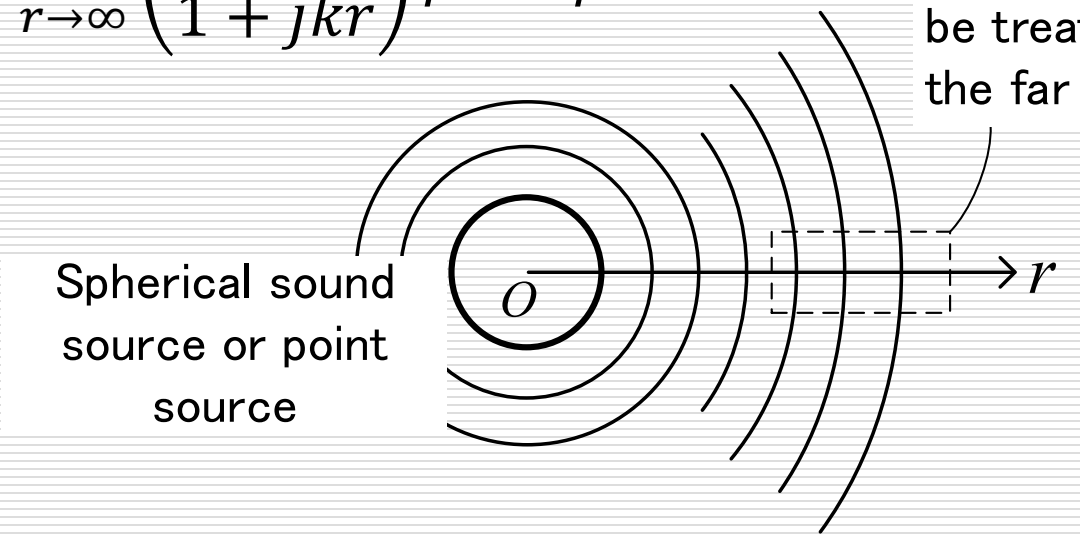
Specific acoustic impedance $Z_s$:

$$Z_s = \frac{p}{v} = \frac{j\omega\rho\Phi}{(1+jkr)\Phi/r} = \frac{j\omega\rho r}{1+jkr} = \rho c \frac{jkr}{1+jkr}$$

The impedance at much far distance ($r \to \infty$).

$$Z_S = \lim_{r\to\infty}\left(\frac{jkr}{1+jkr}\right)\rho c = \rho c$$

The part of propagation wave emitted from point source can be treated as a plane wave in the far field.



# Definition of radiation impedance

Specific acoustic impedance:

$$Z_S = \frac{p}{v} \quad \left(= \frac{sound\ pressure}{particle\ velocity}\right)$$

Acoustic radiation impedance:

$$Z_{rad} = \frac{Sp}{v} \quad \left(= \frac{Force\ (=area \times sound\ pressure)}{particle\ velocity}\right)$$

Radiation impedance is usually defined at the surface of the sound source.

# Radiation impedance of the point source

Pressure on the surface of the source:

$$p|_{r=a} = \rho \left.\frac{\partial \Phi}{\partial t}\right|_{r=a} = j\omega\rho\Phi$$

Particle velocity on the surface of the source:

$$v|_{r=a} = -\left.\frac{\partial \Phi}{\partial r}\right|_{r=a} = \frac{j}{a}\left(\frac{1}{a} + jk\right)\Phi$$

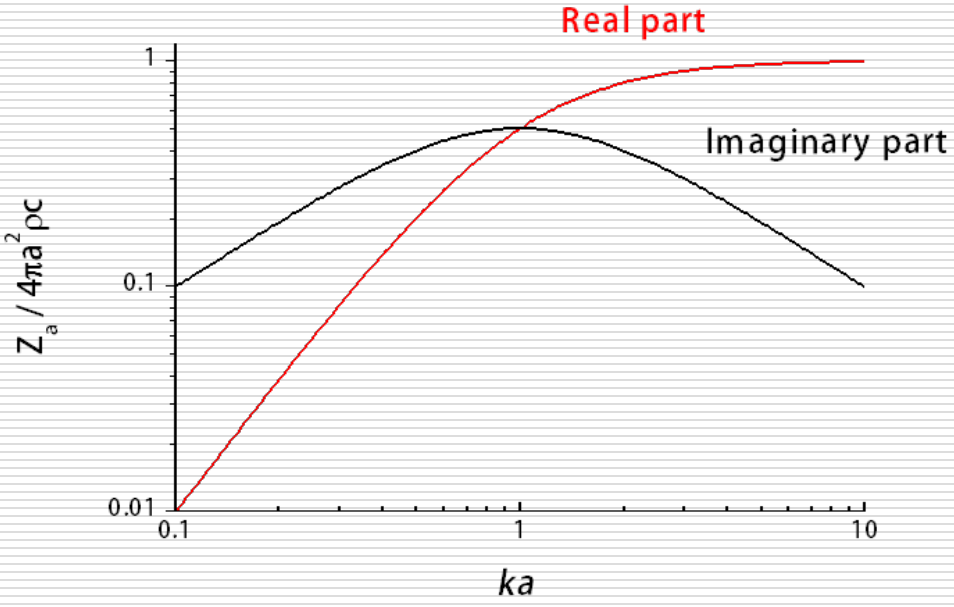
Radiation impedance on the surface:

(area of the surface is $S = 4\pi a^2$)

$$Z_{rad} = \frac{4\pi a^2\, p|_{r=a}}{v|_{r=a}} = 4\pi a^2 \rho c \left(\frac{k^2a^2}{1+k^2a^2} + j\frac{ka}{1+k^2a^2}\right)$$

# Variation of the acoustic radiation impedance

The ratio of the real part and the imaginary part of the acoustic radiation impedance varies against the value of $ka$.



# Meanings of the real part and the imaginary part of the acoustic radiation impedance

$$Z_{rad} = 4\pi a^2 \rho c \left( \underbrace{\frac{k^2 a^2}{1 + k^2 a^2}}_{\text{real part}} + j \underbrace{\frac{ka}{1 + k^2 a^2}}_{\text{imaginary part}} \right)$$

Real part is the component of the emitted sound from the source, that is active impedance.

Imaginary part is the component of the emitted and returned sound to the source, that is reactive impedance.

# Appropriate condition of $ka$ as the sound source

Magnitude of the emitted sound should be larger than that of the sound returned.

Real part should be larger than imaginary part as soon as possible.

$ka$ should be large value.

It is difficult to realize large $ka$ at low frequency.

# Sound field created by piston source (circular plate)

# To estimate the sound field : The Rayleigh integral

Finally, velocity potential by whole planar source is $\Phi_1 + \Phi_2 + \cdots = \sum \Phi_i$

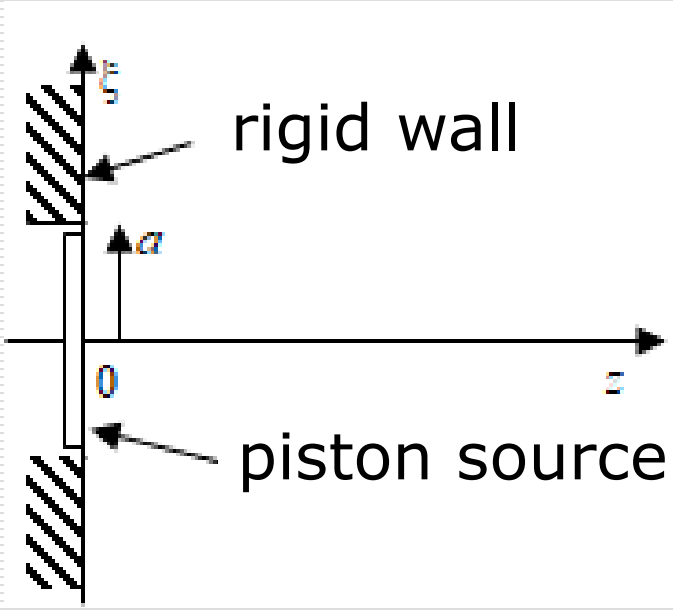
if dS → 0, we obtained the following equation.

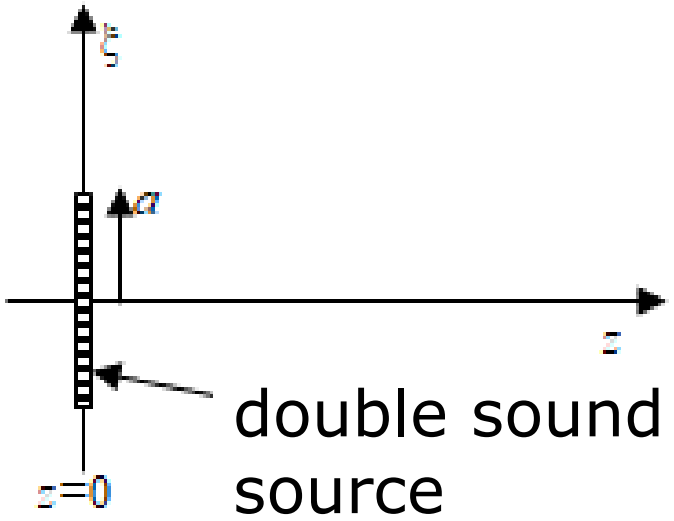
$$\Phi = \frac{V_0}{2\pi} \iint_{S_0} \frac{e^{-jkr}}{r} dS$$

The equation is called “Rayleigh integral”.

# Sound field from piston circular source

Let me consider following piston circular source like speaker.



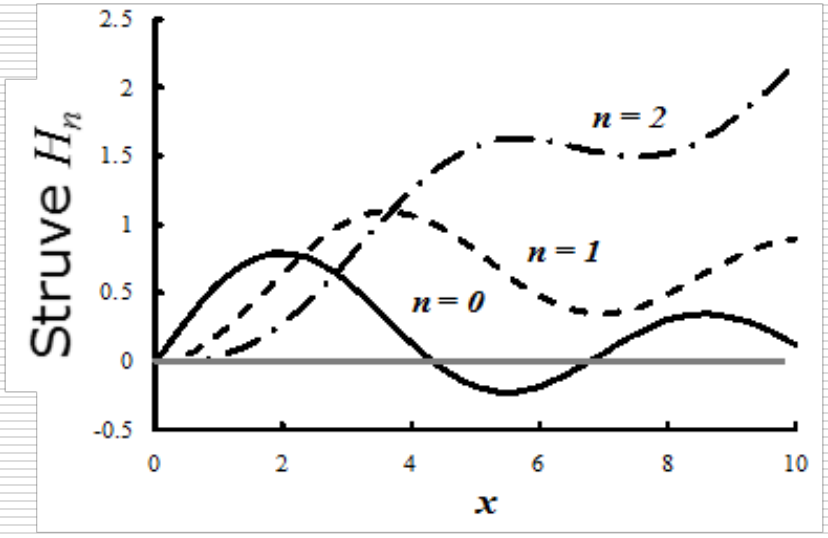
# Radiation impedance of piston source

Radiation impedance of piston is complicated as follows.

$$Z_{rad} = 4\pi a^2 \rho c \left[ \left( 1 - \frac{J_1(2ka)}{ka} \right) + j \frac{H_1(2ka)}{ka} \right]$$

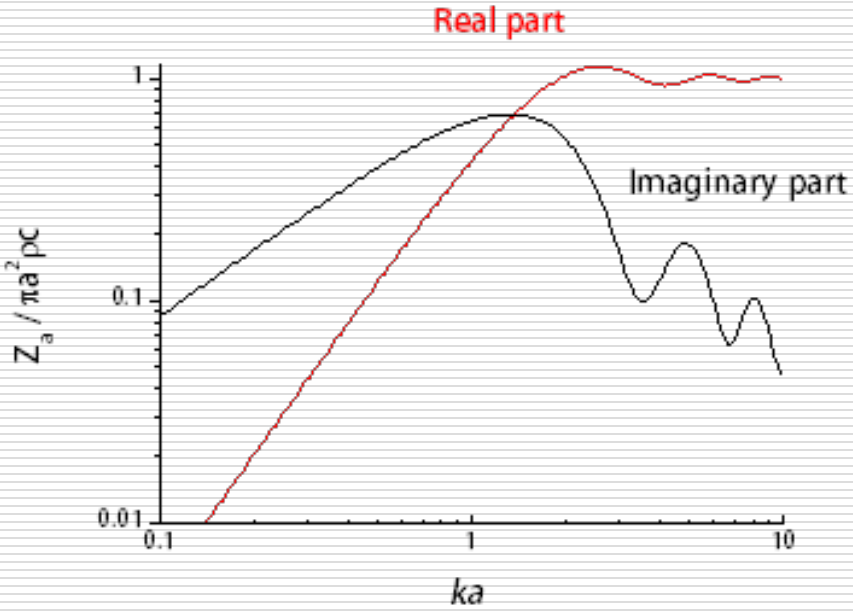
$J_1$ is 1st order Bessel function.

$H_1$ is called as Struve function.



# Variation of radiation impedance of piston source

Variation of the Radiation impedance of $Z_{rad}$ from piston source is almost same as point source.



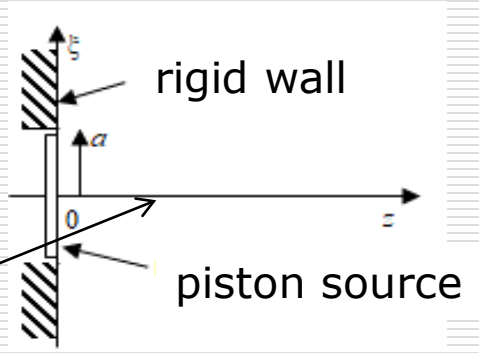
# Question 1

Explain why it is difficult to efficiently emit low-frequency sound using a small loudspeaker.

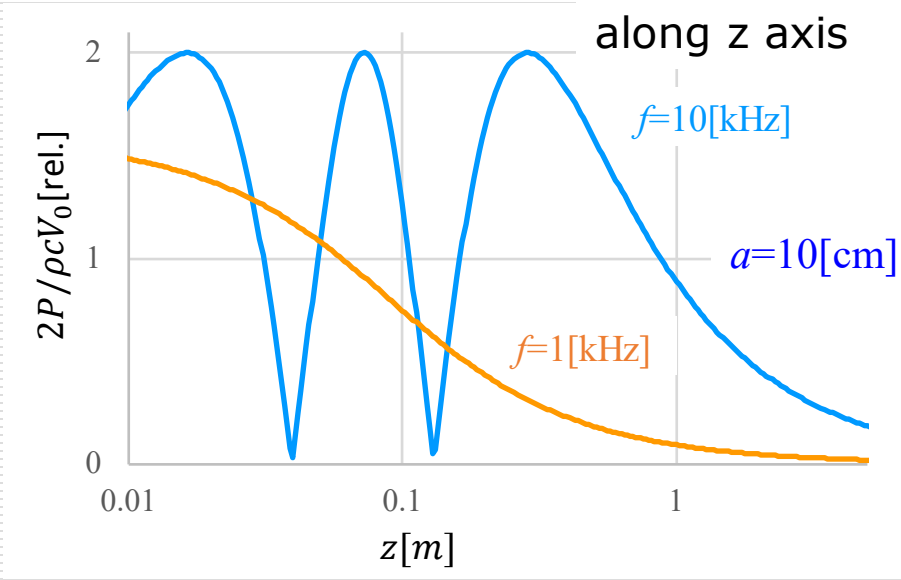
# Sound field along the axis

Sound field along the axis gives the following equation.

$$|P| = 2\rho c V_0 \left| \sin\left[ \frac{k}{2}\left( \sqrt{a^2 + z^2} - z \right) \right] \right|$$




sound field along z axis




## far field

$$z > k a^2 / 2\pi = a^2 / \lambda$$

where $\frac{k}{2}\left( \sqrt{a^2 + z^2} - z \right) < \frac{\pi}{2}$

## near field

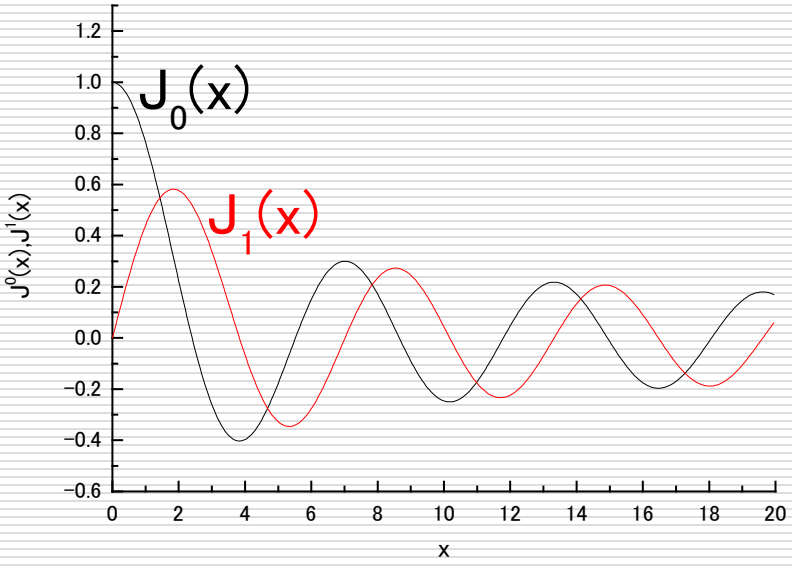
$$z < k a^2 / 2\pi = a^2 / \lambda$$

# Directivity of piston source

Directivity of piston source gives the following equation.

$$D(\theta) = \left| \frac{2 J_1 \left( k a \sin\theta \right)}{k a \sin\theta} \right|$$

Bessel function



# Directivity of piston source

# Common logarithm

In the evaluation or the calculation of large value,  small amount deviation is not important.

Digit order is important in the large value evaluation. 20% or 30% variation is not important.



Logarithmic treatment is useful. Acoustic phenomena is also treated by logarithmic analysis.

$\log_{10} x$ : common logarithm

# dB (decibel)

Relative value is also evaluated by logarithmic analysis. In such a case, dB is also used.

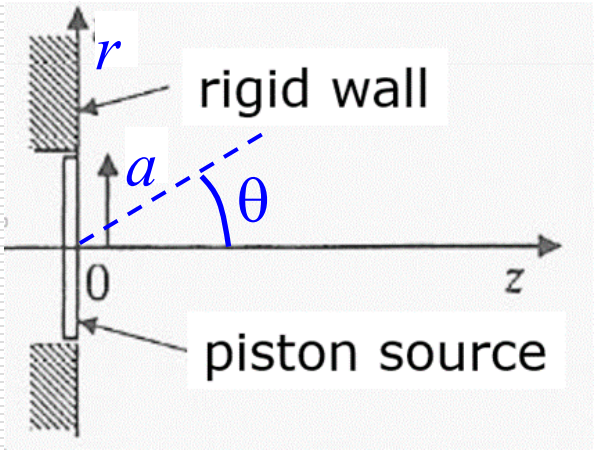
For power evaluation, the output power $P_o$ is compared to the reference value $P_{ref}$ as follows.

$$10 \log_{10} \frac{P_o}{P_{ref}} \text{ [dB]}$$

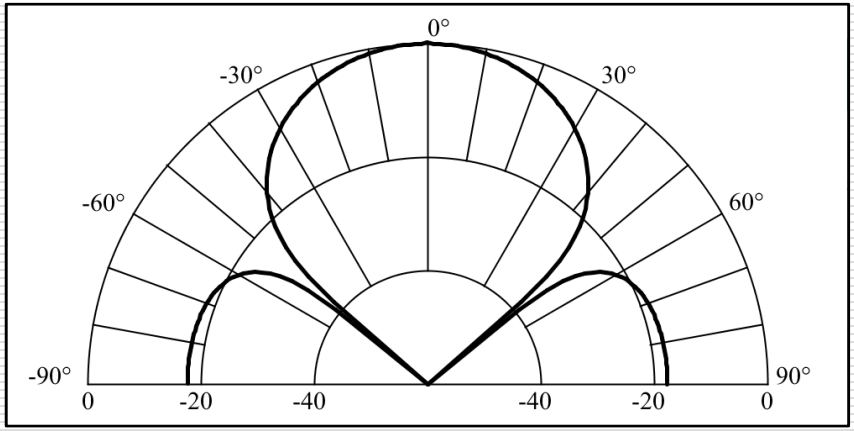
For signal evaluation, the output value $V_o$ is compared to the reference value $V_{ref}$ as follows.

$$20 \log_{10} \frac{V_o}{V_{ref}} \text{ [dB]} \qquad \text{where} \qquad P_o \propto V_o^2$$

# Directivity of piston source



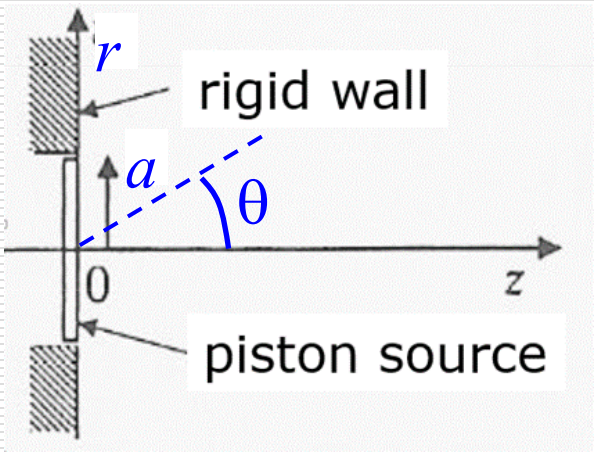

The following directivity pattern shows the sound distribution along θ direction. The maximum value appears in from of the piston source radiation surface (θ =0).




$ka = 5$

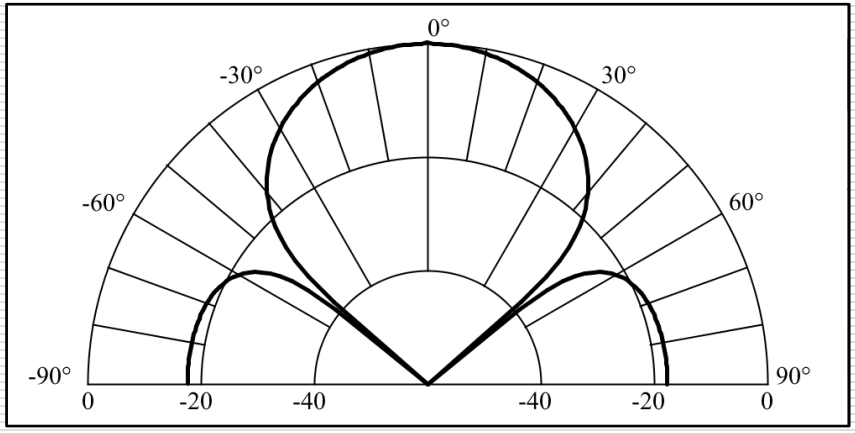
The plotted line is normalized by maximum value.

# Directivity pattern against $ka$

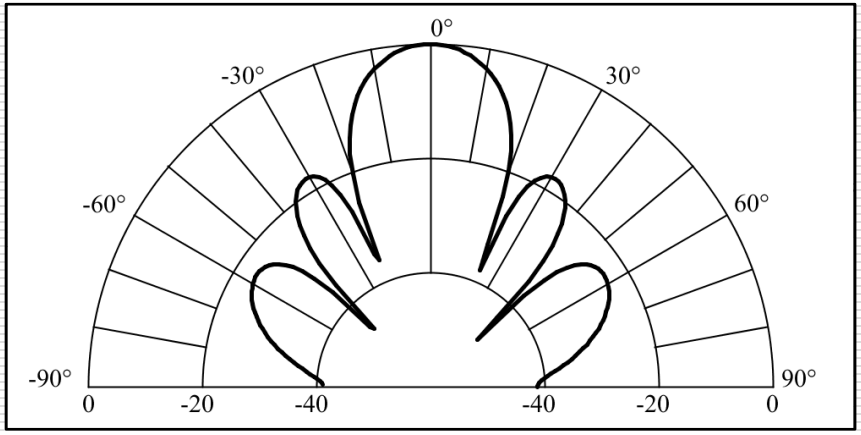



Directivity is sharp at the large $ka$.

Large $ka$ is available at high frequency and large surface.




$k\,a = 5$




$k\,a = 10$

# Question 2

Explain how the directivity pattern emitted by a piston sound source changes with respect to $ka$.

